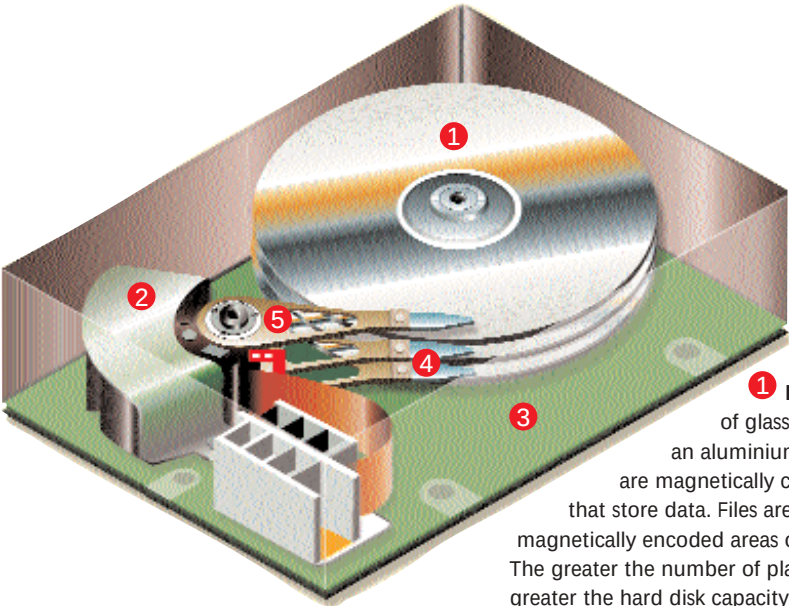




1 Platters: Made of glass substrate or an aluminium alloy, platters are magnetically coated disks that store data. Files are stored as magnetically encoded areas on the platters. The greater the number of platters, the greater the hard disk capacity, though the capacity also depends on the areal density of the platters.

2 Head Actuator: The actuator moves the actuator arms across the platters in order to precisely align the read-write heads above the area storing the data.

3 Logic Card: The printed circuit board at the bottom of the hard disk is called the logic card, and contains a microprocessor and memory. The logic card is responsible for controlling the hard drive. It receives commands via the hard drive controller that is driven by the operating system.

4 Read-Write heads: These are attached to the ends of the actuator arms and are

located on both the top and underside of each hard disk platter. The read-write heads are positioned at a distance of a three millionth of an inch from the platters, and are responsible for reading and writing data.

5 Actuator Arms: Attached to the actuator head at one end, the arms move the read-write heads across the platters to read or write data.

Do Remember

- Prefer a 7,200 rpm drive over a 5,400 rpm, for improved performance—there's hardly any difference now between the prices.
- Never invest in a second-hand hard disk, since it'll be prone to bad sectors and other problems.
- As of today, SATA drives do not provide a significant performance gain over IDE drives, but it is better to buy one if you are upgrading your PC to make it future-proof.

Facts

- The first hard disk, was introduced in 1956. It had a capacity of about 5 MB, stored on 50 24" disks. Its areal density was a mere 2,000 bits per square inch, and its data throughput was 8,800 bits per second.



Buying Tips

- ✓ Speed or space? If you need to store MP3s and DivX films, or to back up heavy image files, buy the largest hard disk you can afford. On the other hand, if you extensively use applications such as image editing or audio processing software, look for the fastest hard disk you can afford—at least a 7,200-rpm drive—and buy the largest in that speed range.
- ✓ As of today, both the SATA and IDE standards provide the same performance, but the future seems to be SATA. So, if your motherboard supports SATA, buy a SATA hard drive.
- ✓ Research the market. Ask your friends or dealers you know, or search the Internet for information about the manufacturer, the reputation they enjoy and their after sales-service.
- ✓ Check the warranty. Apart from the time frame covered, also check the type of warranty. Some manufacturers will replace the hard disk only in the first year, after which they will only repair it. This is rather unreliable, and can lead to more problems in the future.
- ✓ Confirm the time it takes to replace a faulty hard disk. This is especially important if your business relies heavily on data storage. Generally, manufacturers who run a service centre are able to provide quick replacements, while those who don't have service centres could take weeks to replace a faulty hard disk.

Quick Find	
YOU NEED	A steady-going disk that can store all your movies, MP3s and other data with decent performance and a large enough capacity
Basic Use	
LOOK FOR	A 7,200 RPM drive with at least 40 to 80 GB of capacity, with price per MB in the region of 5 to 7 paise.
YOU NEED	A disk with good performance, large storage capacity, fast in copying files on and off the drive. A drive that is future proof and won't become obsolete in the wake of SATA becoming standard
Inter Use	
LOOK FOR	A 7,200 RPM drive with 120 GB to 160 GB of space, and excellent read and write speeds.
YOU NEED	A super-fast drive with excellent performance, low latency, fast in copying data on and off the disk, and also supporting future standards
Prof Use	
LOOK FOR	A drive with at least 8 MB of buffer, spindle speed over 7,200 rpm, extremely low CPU utilisation, good overall performance, and loads of space



The Future

The world of hard disk manufacturers is changing—on one side, the demand for storage is on the rise, while on the other side the demand for faster drives is also rising. There's pressure from existing users for better performance, as well as pressure due to the emergence of technologies such as SATA (Serial ATA). SATA is the next-generation revolution in hard disk interfaces. It is one standard that's causing a big impact in the internal interface industry. The Serial ATA interface helps transfer data at a speed of 150 MBps with ease, and is moving on to pick-up speeds of up to 300 MBps by 2005 and 600 MBps by 2010.